

The hands-on workshop (Hagenberg 04.05.2026), is devoted to training neural networks for recognizing handwritten digits using a simple low-res data set ( $6 \times 4$ ).

**Exercise 1 (30 points)** Start from the notebook `ANN_Basics.ipynb` and perform the following steps:

1. Try at least 10 different configurations of traditional networks (only sigmoid activations in hidden layers, no dropout, stochastic gradient descent; modify only layers, learning rate, and number of epochs).
2. Try at least 10 different configurations of modern networks (any activation in hidden layers allowed, dropout allowed, any optimization algorithm allowed).

*Document your settings and the resulting validation accuracies carefully. Discuss and interpret your results. Please hand in your notebook (including discussion of results) and an HTML dump of it.*